

Identity Non-Conservation Under Admissible Propagation: Neutrino Oscillation as a Structural Necessity

Amos Jay Maley

Abstract

Neutrino oscillation is conventionally presented as a dynamical consequence of mass-flavor misalignment encoded in the PMNS matrix. While empirically correct, this presentation obscures a deeper structural necessity. In this paper, we show that neutrino oscillation is required by admissibility under standing conservation once interaction-defined identity labels fail to commute with irreversible propagation. Using the Mealy admissibility constraint, we demonstrate that fixed flavor identity is forbidden in neutrinos, and that oscillation arises as the admissible form of identity drift rather than as a contingent dynamical anomaly. No new dynamics are proposed; the result is classificatory and eliminative, reframing oscillation as a consequence of reference stability rather than mass mixing ontology.

1 Introduction

Neutrino oscillation is an experimentally established phenomenon in which neutrinos produced in one interaction-defined flavor are later detected as another. Standard treatments describe this behavior using mass eigenstates, mixing matrices, and phase evolution. While successful, such treatments implicitly elevate bookkeeping constructs into ontic descriptors.

This paper advances a narrower and structurally grounded claim:

Neutrino oscillation is not merely compatible with admissible propagation—it is required once admissibility under standing conservation is enforced.

The necessity arises not from new dynamics, but from the inadmissibility of fixed identity labels under irreversible propagation when those labels are defined contextually by interaction.

2 Identity, Propagation, and Admissibility

2.1 Interaction-defined identity

Neutrino flavor is defined operationally by weak interaction vertices. It is not an intrinsic propagation invariant but a contextual label assigned at emission or detection. Flavor therefore functions as a bookkeeping identity rather than an ontic one.

2.2 Irreversible propagation

Neutrino propagation involves phase accumulation over spacetime intervals that cannot be undone or globally reindexed. This evolution is irreversible in the admissibility sense: later descriptions cannot erase earlier distinctions without loss of standing.

2.3 The admissibility constraint (Mealy Theorem)

The Mealy Theorem imposes the following structural requirement:

A system preserving reference under irreversible evolution may not maintain latent internal state distinctions that affect admissible outcomes unless those distinctions are fixed at the point of reference.

Any identity label that is not fixed at emission, evolves nontrivially during propagation, and re-enters admissible interaction outcomes cannot be treated as ontically conserved.

3 Failure of Flavor Conservation as a Structural Necessity

Assume, for contradiction, that neutrino flavor is conserved.

Then flavor would constitute an ontic invariant persisting through propagation. Phase evolution would be required to commute with flavor labeling, and detection outcomes would be re-expressible without identity drift.

This assumption fails. Propagation occurs in a basis not aligned with interaction labeling. Phase accumulation produces interference effects that alter admissible detection outcomes. Fixed flavor identity would therefore require reassigning outcomes without admissible fixation, violating standing.

The contradiction is structural rather than empirical.

4 Oscillation as Admissible Identity Drift

It is essential to distinguish the present claim from the mere observation that flavor and propagation bases differ. Basis mismatch alone is a representational fact and does not, by itself, impose necessity. The necessity of oscillation arises only once such mismatch is combined with admissibility under irreversible propagation, which forbids fixed identity labels that fail to commute with admissible evolution.

Given this constraint, only one admissible resolution exists: identity must be allowed to drift relationally under propagation.

Neutrino oscillation is precisely this drift. Identity is not preserved as a primitive, but is re-constituted contextually at interaction. Intermediate propagation carries no fixed flavor standing. Oscillation is therefore not a failure of conservation laws, but a successful enforcement of admissibility.

5 Why Neutrinos Are Special

Charged leptons do not exhibit analogous oscillation because their interaction and propagation bases coincide to high accuracy. Identity labels commute with admissible evolution, and no latent internal distinction survives unfixed.

Neutrinos differ because their interaction-defined identity is weakly coupled, their propagation basis is effectively hidden, and the Mealy constraint becomes active. Oscillation is therefore not a general particle feature but a structural consequence of weakly anchored identity. The argument is agnostic to whether neutrinos are Dirac or Majorana.

6 Scope and Non-Derivation

The present analysis does not derive oscillation frequencies, mixing angles, mass splittings, or CP-violating phases. These quantities remain contingent bookkeeping parameters within the Standard Model.

Non-Derivation Statement. This result concerns only the structural inadmissibility of fixed flavor identity under irreversible propagation. In neutrinos, this prohibition is realized empirically as oscillation, independent of the specific values of mixing parameters.

7 Classification Within the Maley Corpus

This result is classified as Family II (Applied / Derived). It is non-dynamical, non-constructive, and eliminative with respect to flavor realism. It introduces no new primitives or enforcement tools and depends on admissibility, standing conservation, and the Mealy Theorem.

8 Conclusion

Neutrino oscillation is best understood as the admissible resolution of identity non-conservation under irreversible propagation. Once interaction-defined labels fail to commute with admissible evolution, fixed identity conservation is forbidden. In neutrinos, this prohibition is structurally forced to appear as oscillation.